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Multi-Asset Portfolio Construction with Hybrid CVaR-QAOA

Naga Snigdha Kondepoti & Dasari Ravi Teja

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1 Introduction

Multi-asset portfolio construction is a fundamental optimization problem in quantitative finance that seeks to allocate capital across diverse asset classes to maximize expected returns while managing risks and satisfying investment constraints. As the number of assets and investment constraints increases, the portfolio optimization problem becomes computationally complex due to its combinatorial nature.

2 Problem Statement

Given return assumptions, risk estimates, cost estimates and portfolio constraints of various asset classes model a portfolio using quantum computing that improves expected utility while honouring goals, guardrails and implementation cost.

3 Mathematical Formulation

This section defines the mathematical formulation of the multi-asset portfolio construction problem focusing on continuous & binary decision variables, linear constraints and quadratic objectives. Then the constraint optimization problem (or the cost function) is defined. This constraint optimization problem is converted to unconstrained optimization problem so that it is compatible with a quantum optimization algorithm.

First, the model inputs and tunable investment goals are introduced, followed by the decision variables and practical portfolio constraints for the multi-asset portfolio construction problem. The classical mean-variance optimization model is formulated and then extended to incorporate tunable goals (user-inputs) and real-world constraints. Finally, the constrained optimization problem is transformed into a Quadratic Unconstrained Binary Optimization (QUBO) formulation suitable for quantum optimization.

3.1 Model Inputs

Consider a portfolio consisting of N assets belonging to different asset classes. For each asset consider the following financial inputs:

- Expected return vector

$$\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_N]^T$$

- Covariance matrix of asset returns

$$\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$$

- Income yield vector

$$\mathbf{y} = [y_1, y_2, \dots, y_N]^T$$

- Drawdown risk score vector

$$\mathbf{d} = [d_1, d_2, \dots, d_N]^T$$

- Transaction cost vector

$$\mathbf{c} = [c_1, c_2, \dots, c_N]^T$$

3.2 Tunable Investment Goals

To allow the investors to adjust the portfolio according to different investment preferences, 4 tunable parameters are introduced to control the relative importance of each objective:

1. α : Growth preference
2. β : Income preference
3. γ : Drawdown control
4. δ : Cost sensitivity

By adjusting these parameters, investors can prioritize capital appreciation, income generation, downside protection or transaction cost minimization.

3.3 Decision Variables

Two types of decision variables are used in the portfolio optimization model.

For each asset define a continuous portfolio weight

$$0 \leq w_i \leq 1,$$

where w_i represents the proportion of total capital allocated to asset i .

Collect the continuous portfolio weights as

$$\mathbf{w} = [w_1, w_2, \dots, w_N]^T,$$

For each asset define a binary decision variable as

$$x_i = \begin{cases} 1, & \text{if asset } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Collect the binary decisions as

$$\mathbf{x} = [x_1, x_2, \dots, x_N]^T,$$

The quantum optimization algorithms based on the Quadratic Unconstrained Binary Optimization (QUBO) formulation require the objective function to be expressed solely in terms of binary decision variables. Furthermore, current quantum hardware has limited qubit counts and is susceptible to noise, making it difficult to directly optimize continuous variables.

To make the optimization problem tractable on current quantum devices, each continuous portfolio weight is mapped to its corresponding binary selection variable through a proportional relationship

$$w_i = a_i x_i, \tag{1}$$

or, in vector form,

$$\mathbf{w} = A\mathbf{x}, \tag{2}$$

where

$$A = \text{diag}(a_1, a_2, \dots, a_N),$$

and a_i is an allocation coefficient associated with asset i .

The coefficient a_i specifies the allocation assigned to asset i whenever $x_i = 1$; otherwise, the allocation is zero.

3.4 Portfolio Constraints

The optimized portfolio is required to satisfy practical investment constraints. These constraints ensure that the resulting allocation remains feasible and consistent with investment policies.

We consider 2 equality constraints and 2 inequality constraints.

3.4.1 Equality Constraints

- **Budget Constraint** The total portfolio weight allocation must equal to 1

$$\sum_{i=1}^N w_i = 1.$$

- **Cardinality Constraint**

The number of selected assets is limited by

$$\sum_{i=1}^N x_i = K,$$

where $K \leq N$ denotes the maximum allowable number of assets.

3.4.2 Inequality Constraints

- **Asset Allocation Bounds**

Each selected asset must satisfy

$$l_i x_i \leq w_i \leq u_i x_i,$$

where l_i and u_i denote the minimum and maximum allowable allocation for asset i .

- **Sector Exposure Constraint**

To prevent excessive concentration, the allocation within each sector must satisfy

$$\sum_{i \in S_j} w_i \leq b_j,$$

where S_j represents sector j and b_j is the maximum allowable exposure to that sector.

3.5 Portfolio Penalties

To transform the constrained optimization problem into a form suitable for quantum optimization, each constraint is incorporated into the objective function through quadratic penalty terms.

3.5.1 Penalty: Equality Constraints

- **Penalty 1: Budget Constraint**

The budget constraint is

$$\sum_{i=1}^N w_i = 1. \tag{3}$$

The corresponding quadratic penalty is

$$P_{\text{budget}} = \eta_1 \left(\sum_{i=1}^N w_i - 1 \right)^2 \quad (4)$$

where $\eta_1 > 0$ is the penalty coefficient.

- **Penalty 2: Cardinality Constraint**

The cardinality constraint is

$$\sum_{i=1}^N x_i = K. \quad (5)$$

Its quadratic penalty is written as

$$P_{\text{card}} = \eta_2 \left(\sum_{i=1}^N x_i - K \right)^2 \quad (6)$$

where $\eta_2 > 0$ controls the penalty strength.

3.5.2 Penalty: Inequality Constraints

Unlike equality constraints, inequality constraints are only violated when the feasible region is exceeded. Rather than introducing slack variables to convert the inequalities into equalities, we adopt an exterior quadratic penalty approach, as commonly employed in constrained optimization and recent quantum optimization formulations [[1], [2], [3]]. In this approach, a penalty is imposed only when an inequality constraint is violated, while feasible solutions incur zero penalty.

For a general inequality constraint

$$g_k(x) \leq 0,$$

the corresponding penalty function is defined as

$$P_k(x) = \max(0, g_k(x))^2.$$

This formulation preserves the original inequality constraints without increasing the number of optimization variables, thereby avoiding the additional complexity associated with slack-variable-based reformulations.

- **Penalty 3: Asset Allocation Bounds**

The asset allocation bounds are given by

$$l_i x_i \leq w_i \leq u_i x_i.$$

These inequalities are written in the standard form

$$l_i x_i - w_i \leq 0,$$

and

$$w_i - u_i x_i \leq 0.$$

The corresponding penalty function is

$$P_{\text{bound}} = \eta_3 \sum_{i=1}^N \left[\max(0, l_i x_i - w_i)^2 + \max(0, w_i - u_i x_i)^2 \right] \quad (7)$$

where $\eta_3 > 0$ is the penalty coefficient controlling the severity of violations of the allocation bounds.

- **Penalty 4: Sector Exposure Constraint**

The sector exposure constraint is

$$\sum_{i \in S_j} w_i \leq b_j,$$

for each sector S_j .

Writing the constraint in the standard inequality form gives

$$\sum_{i \in S_j} w_i - b_j \leq 0.$$

The corresponding exterior quadratic penalty is

$$P_{\text{sector}} = \eta_4 \sum_j \max \left(0, \sum_{i \in S_j} w_i - b_j \right)^2 \quad (8)$$

where $\eta_4 > 0$ is the penalty coefficient associated with the sector exposure constraint.

3.5.3 Total Penalty

The overall penalty function is obtained by combining the penalty terms corresponding to both the equality and inequality constraints.

First, the equality constraint penalty function is defined as

$$P_{\text{eq}}(\mathbf{w}, \mathbf{x}) = P_{\text{budget}} + P_{\text{card}}. \quad (9)$$

Substituting (4) and (6) gives

$$P_{\text{eq}}(\mathbf{w}, \mathbf{x}) = \eta_1 \left(\sum_{i=1}^N w_i - 1 \right)^2 + \eta_2 \left(\sum_{i=1}^N x_i - K \right)^2. \quad (10)$$

Similarly, the inequality constraint penalty function is defined as

$$P_{\text{ineq}}(\mathbf{w}, \mathbf{x}) = P_{\text{bound}} + P_{\text{sector}}. \quad (11)$$

Substituting (7) and (8) yields

$$\begin{aligned} P_{\text{ineq}}(\mathbf{w}, \mathbf{x}) = & \eta_3 \sum_{i=1}^N \left[\max(0, l_i x_i - w_i)^2 + \max(0, w_i - u_i x_i)^2 \right] \\ & + \eta_4 \sum_j \max \left(0, \sum_{i \in S_j} w_i - b_j \right)^2. \end{aligned} \quad (12)$$

Therefore, the complete penalty function used in the unconstrained optimization formulation is obtained by combining the equality and inequality penalty functions,

$$P(\mathbf{w}, \mathbf{x}) = P_{\text{eq}}(\mathbf{w}, \mathbf{x}) + P_{\text{ineq}}(\mathbf{w}, \mathbf{x}) \quad (13)$$

The resulting penalty function preserves the original constraint structure while transforming the constrained portfolio optimization problem into an unconstrained formulation. Equality constraints are enforced through quadratic penalties, whereas inequality constraints are penalized only when violated using exterior quadratic penalty functions. This formulation is subsequently incorporated into the objective function to obtain the unconstrained optimization problem suitable for QUBO construction and Hamiltonian mapping.

3.6 Baseline Mean–Variance Optimization

The classical Markowitz portfolio optimization model considers the trade-off between expected portfolio return and portfolio risk. The optimization problem is formulated as

$$\min_{\mathbf{w}} -\boldsymbol{\mu}^T \mathbf{w} + \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}, \quad (14)$$

where $\lambda > 0$ denotes the investor’s risk-aversion coefficient. The first term represents the expected portfolio return, whereas the second term penalizes portfolio variance. Increasing the value of λ produces more conservative portfolios with lower risk.

3.7 Multi-Objective Portfolio Optimization

Although the classical mean–variance model considers expected return and portfolio risk, practical portfolio construction often requires multiple investment objectives. Besides capital appreciation, investors may also seek regular income, reduced downside risk and lower implementation costs.

Using the financial inputs and tunable investment parameters introduced in Section 3.1 and Section 3.2, the portfolio objective is extended as

$$\begin{aligned} \min_{\mathbf{w}} \quad & -\alpha \boldsymbol{\mu}^T \mathbf{w} - \beta \mathbf{y}^T \mathbf{w} \\ & + \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} + \gamma \mathbf{d}^T \mathbf{w} + \delta \mathbf{c}^T \mathbf{w} \end{aligned} \quad (15)$$

where

- α controls the emphasis on expected capital appreciation,
- β controls the emphasis on portfolio income,
- λ denotes the risk-aversion coefficient,
- γ controls the drawdown penalty,
- δ controls the transaction-cost penalty.

The proposed formulation enables investors to generate portfolios that balance growth, income, downside protection and implementation cost according to their investment preferences.

3.8 Mixed-Integer Quadratic Programming (MIQP) Formulation

The optimization model in Equation (15) is subject to the portfolio constraints introduced in Section 3.4. Since the formulation contains continuous portfolio weights together with binary asset-selection variables, the resulting optimization problem is a Mixed-Integer Quadratic Programming (MIQP) problem.

The MIQP formulation is given by

$$\begin{aligned} \min_{\mathbf{w}, \mathbf{x}} \quad & -\alpha \boldsymbol{\mu}^T \mathbf{w} - \beta \mathbf{y}^T \mathbf{w} + \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} + \gamma \mathbf{d}^T \mathbf{w} + \delta \mathbf{c}^T \mathbf{w} \\ \text{subject to} \quad & \sum_{i=1}^N w_i = 1, \\ & \sum_{i=1}^N x_i = K, \\ & l_i x_i \leq w_i \leq u_i x_i, \\ & \sum_{i \in S_j} w_i \leq b_j. \end{aligned} \quad (16)$$

The presence of binary decision variables together with a quadratic objective function makes the optimization problem NP-hard for large portfolios, motivating the use of quantum optimization techniques.

3.9 Unconstrained Optimization Formulation

The constrained Mixed-Integer Quadratic Programming (MIQP) problem in Equation (16) is transformed into an unconstrained binary optimization problem by incorporating the portfolio constraints into the objective function through penalty terms.

We first define the original portfolio objective as

$$C(\mathbf{w}) = -\alpha \boldsymbol{\mu}^T \mathbf{w} - \beta \mathbf{y}^T \mathbf{w} + \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} + \gamma \mathbf{d}^T \mathbf{w} + \delta \mathbf{c}^T \mathbf{w}. \quad (17)$$

The total penalty function is given by Equation (9), where the equality and inequality penalty terms are incorporated into the objective function.

Therefore, the penalized constrained objective can be written as

$$F(\mathbf{w}, \mathbf{x}) = C(\mathbf{w}) + P(\mathbf{w}, \mathbf{x}). \quad (18)$$

The objective in Equation (18) still contains the continuous portfolio weights $\mathbf{w}$. To obtain an objective expressed solely in terms of binary variables, the equal-weight portfolio assumption introduced through the binary mapping is now applied.

3.9.1 Equal-Weight Portfolio Assumption

As established previously, the portfolio weight vector is related to the binary asset-selection vector through the linear mapping

$$\mathbf{w} = A\mathbf{x}, \quad (19)$$

where A is a diagonal scaling matrix. For the equal-weight portfolio considered in this work, exactly K assets are selected and each selected asset is assigned an identical allocation of $1/K$. Consequently, the mapping matrix is chosen as

$$A = \frac{1}{K} I, \quad (20)$$

where I denotes the $N \times N$ identity matrix. Substituting (20) into (19) gives

$$\mathbf{w} = \frac{1}{K} \mathbf{x}. \quad (21)$$

Equivalently, for each asset,

$$w_i = \frac{x_i}{K}, \quad i = 1, \dots, N, \quad (22)$$

where $x_i = 1$ indicates that asset i is selected and therefore receives an allocation of $1/K$, while $x_i = 0$ indicates that asset i is not selected and receives zero allocation.

The total portfolio weight is therefore

$$\sum_{i=1}^N w_i = \sum_{i=1}^N \frac{x_i}{K} = \frac{1}{K} \sum_{i=1}^N x_i. \quad (23)$$

From the previously defined cardinality constraint,

$$\sum_{i=1}^N x_i = K, \quad (24)$$

substituting (24) into (23) yields

$$\sum_{i=1}^N w_i = \frac{1}{K} K = 1. \quad (25)$$

Therefore, under the equal-weight assumption, the budget constraint is automatically satisfied whenever the cardinality constraint is enforced. Consequently, the budget equality penalty is redundant and may be omitted from the QUBO formulation, while only the cardinality equality penalty is retained.

3.9.2 Transformation of the Objective

Substituting

$$\mathbf{w} = \frac{1}{K} \mathbf{x}$$

into Equation (17) gives the objective entirely in terms of the binary variables:

$$\begin{aligned} C(\mathbf{x}) = & -\frac{\alpha}{K} \boldsymbol{\mu}^T \mathbf{x} - \frac{\beta}{K} \mathbf{y}^T \mathbf{x} + \frac{\lambda}{K^2} \mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} \\ & + \frac{\gamma}{K} \mathbf{d}^T \mathbf{x} + \frac{\delta}{K} \mathbf{c}^T \mathbf{x}. \end{aligned} \quad (26)$$

Similarly, after applying the equal-weight mapping, the total penalty function becomes

$$P(\mathbf{x}) = P_{\text{card}}(\mathbf{x}) + P_{\text{ineq}}(\mathbf{x}), \quad (27)$$

Therefore, the unconstrained binary objective is

$$F(\mathbf{x}) = C(\mathbf{x}) + P_{\text{card}}(\mathbf{x}) + P_{\text{ineq}}(\mathbf{x}). \quad (28)$$

Substituting Equation (26) and Equation (6) into Equation (28) gives

$$\begin{aligned} F(\mathbf{x}) = & -\frac{\alpha}{K} \boldsymbol{\mu}^T \mathbf{x} - \frac{\beta}{K} \mathbf{y}^T \mathbf{x} + \frac{\lambda}{K^2} \mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} \\ & + \frac{\gamma}{K} \mathbf{d}^T \mathbf{x} + \frac{\delta}{K} \mathbf{c}^T \mathbf{x} \\ & + \eta_2 \left(\sum_{i=1}^N x_i - K \right)^2 + P_{\text{ineq}}(\mathbf{x}). \end{aligned} \quad (29)$$

where

$$\begin{aligned} P_{\text{ineq}}(\mathbf{x}) = & \eta_3 \sum_{i=1}^N \left[\max \left(0, \left(l_i - \frac{1}{K} \right) x_i \right)^2 + \max \left(0, \left(\frac{1}{K} - u_i \right) x_i \right)^2 \right] \\ & + \eta_4 \sum_j \max \left(0, \frac{1}{K} \sum_{i \in S_j} x_i - b_j \right)^2. \end{aligned} \quad (30)$$

Equation (29) is expressed entirely in terms of binary decision variables and therefore provides the starting point for the subsequent QUBO construction.

3.9.3 Expansion into Quadratic, Linear and Constant Terms

To obtain the explicit QUBO structure, the quadratic covariance term and the cardinality penalty are expanded separately.

First, the covariance term can be written as

$$\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} = \sum_{i=1}^N \sigma_{ii} x_i^2 + 2 \sum_{i < j} \sigma_{ij} x_i x_j. \quad (31)$$

Since the decision variables are binary,

$$x_i^2 = x_i. \quad (32)$$

Therefore,

$$\mathbf{x}^T \Sigma \mathbf{x} = \sum_{i=1}^N \sigma_{ii} x_i + 2 \sum_{i < j} \sigma_{ij} x_i x_j. \quad (33)$$

The cardinality penalty is expanded as

$$\begin{aligned} P_{\text{card}}(\mathbf{x}) &= \eta_2 \left(\sum_{i=1}^N x_i - K \right)^2 \\ &= \eta_2 \left[\left(\sum_{i=1}^N x_i \right)^2 - 2K \sum_{i=1}^N x_i + K^2 \right]. \end{aligned} \quad (34)$$

Using $x_i^2 = x_i$,

$$\left(\sum_{i=1}^N x_i \right)^2 = \sum_{i=1}^N x_i + 2 \sum_{i < j} x_i x_j. \quad (35)$$

Hence,

$$P_{\text{card}}(\mathbf{x}) = 2\eta_2 \sum_{i < j} x_i x_j + \eta_2(1 - 2K) \sum_{i=1}^N x_i + \eta_2 K^2. \quad (36)$$

Substituting the covariance expansion Equation (33) and the cardinality expansion Equation (36) into Equation (29), the unconstrained binary objective can be decomposed into quadratic, linear, constant and inequality penalty components:

$$\boxed{F(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0 + P_{\text{ineq}}(\mathbf{x})}. \quad (37)$$

The quadratic component is

$$\boxed{Q(\mathbf{x}) = \frac{2\lambda}{K^2} \sum_{i < j} \sigma_{ij} x_i x_j + 2\eta_2 \sum_{i < j} x_i x_j}. \quad (38)$$

The linear component is

$$\boxed{L(\mathbf{x}) = \sum_{i=1}^N \left[-\frac{\alpha \mu_i}{K} - \frac{\beta y_i}{K} + \frac{\lambda \sigma_{ii}}{K^2} + \frac{\gamma d_i}{K} + \frac{\delta c_i}{K} + \eta_2(1 - 2K) \right] x_i}. \quad (39)$$

The constant component is

$$\boxed{C_0 = \eta_2 K^2}. \quad (40)$$

The constant term C_0 does not affect the location of the optimum and can therefore be omitted when constructing the QUBO matrix.

The resulting formulation can therefore be summarized as

$$\boxed{F(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0 + P_{\text{ineq}}(\mathbf{x})}. \quad (41)$$

The first three terms constitute the quadratic polynomial arising from the portfolio objective and cardinality constraint. The remaining term $P_{\text{ineq}}(\mathbf{x})$ represents penalties associated with violations of the inequality constraints. The treatment of these inequality penalties is considered separately in the subsequent QUBO construction.

4 Hybrid Quantum–Classical Optimization Framework

The unconstrained binary objective derived in the previous section can be written as

$$F(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0 + P_{\text{ineq}}(\mathbf{x}), \quad (42)$$

where $Q(\mathbf{x})$ and $L(\mathbf{x})$ denote the quadratic and linear components of the portfolio objective together with the equality-constraint penalty, C_0 is a constant term, and $P_{\text{ineq}}(\mathbf{x})$ represents the exterior penalty associated with inequality constraint violations.

Conventional QUBO formulations incorporate every constraint directly into the quantum cost Hamiltonian by converting all constraints into quadratic binary penalty terms. For inequality constraints, this typically requires additional auxiliary or slack variables, increasing both the number of binary variables and the complexity of the resulting Hamiltonian.

In this work, a different hybrid optimization strategy is adopted. Rather than embedding the inequality constraints into the quantum Hamiltonian, the quantum optimization stage is restricted to the unconstrained quadratic objective,

$$F_Q(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0, \quad (43)$$

while the inequality penalty term,

$$P_{\text{ineq}}(\mathbf{x}),$$

is evaluated only during the subsequent classical post-processing stage.

This design avoids introducing slack variables, preserves the original inequality constraints, and produces a substantially simpler Hamiltonian for quantum optimization. The quantum computer is therefore responsible for exploring the combinatorial asset-selection space, whereas feasibility assessment and continuous portfolio-weight optimization are performed classically.

The proposed optimization framework consists of four sequential stages:

1. Construction of an unconstrained Ising Hamiltonian from $F_Q(\mathbf{x})$;
2. Optimization of the Hamiltonian using the Conditional Value-at-Risk Quantum Approximate Optimization Algorithm (CVaR-QAOA);
3. Classical evaluation of the complete objective function, including the inequality penalties, for every sampled bit string;
4. Continuous portfolio-weight optimization for the feasible candidate portfolios generated by the quantum stage.

Consequently, the proposed methodology forms a hybrid quantum–classical optimization pipeline in which the quantum processor performs a global combinatorial search over binary asset selections, while the classical optimizer computes the final investment weights satisfying the original portfolio constraints.

4.1 Construction of the Quantum Cost Hamiltonian

Since the inequality penalties are intentionally excluded from the quantum optimization stage, only the unconstrained binary objective in Equation (62) is mapped into an Ising Hamiltonian.

The quantum objective can be expressed in the standard QUBO form as

$$F_Q(\mathbf{x}) = \sum_{i < j} q_{ij} x_i x_j + \sum_{i=1}^N q_i x_i + C_0, \quad (44)$$

where q_{ij} denotes the quadratic interaction coefficient between binary variables x_i and x_j , while q_i denotes the corresponding linear coefficient.

For the proposed portfolio optimization model,

$$q_{ij} = \frac{2\lambda}{K^2} \sigma_{ij} + 2\eta_2, \quad i < j, \quad (45)$$

and

$$q_i = -\frac{\alpha\mu_i}{K} - \frac{\beta y_i}{K} + \frac{\lambda\sigma_{ii}}{K^2} + \frac{\gamma d_i}{K} + \frac{\delta c_i}{K} + \eta_2(1 - 2K), \quad (46)$$

with constant term

$$C_0 = \eta_2 K^2. \quad (47)$$

The corresponding symmetric QUBO matrix is defined as

$$F_Q(\mathbf{x}) = \mathbf{x}^T \mathbf{Q} \mathbf{x} + C_0, \quad (48)$$

where

$$Q_{ii} = q_i, \quad Q_{ij} = Q_{ji} = \frac{q_{ij}}{2}, \quad i \neq j. \quad (49)$$

The factor of one-half in the off-diagonal elements guarantees that

$$\mathbf{x}^T \mathbf{Q} \mathbf{x} = \sum_i q_i x_i + \sum_{i < j} q_{ij} x_i x_j,$$

using the binary property

$$x_i^2 = x_i.$$

Unlike conventional QUBO formulations, no additional quadratic terms arising from the inequality penalties are included in the Hamiltonian construction. Consequently, the quantum optimization problem contains only the interactions originating from the portfolio objective and the cardinality constraint, while all inequality constraints remain part of the classical optimization stage.

4.2 Binary-to-Ising Transformation

The QUBO formulation in Equation (44) is transformed into an Ising model to obtain a Hamiltonian that can be implemented within the Quantum Approximate Optimization Algorithm (QAOA). This transformation maps each binary decision variable to the eigenvalues of a Pauli-Z operator acting on a qubit.

The binary variables

$$x_i \in \{0, 1\}$$

are mapped to spin variables

$$z_i \in \{-1, +1\}$$

through the standard affine transformation

$$x_i = \frac{1 - z_i}{2}, \quad (50)$$

or equivalently,

$$z_i = 1 - 2x_i. \quad (51)$$

Substituting Equation (50) into the QUBO objective yields

$$\begin{aligned}
F_Q(\mathbf{z}) &= \sum_{i < j} q_{ij} \left(\frac{1 - z_i}{2} \right) \left(\frac{1 - z_j}{2} \right) \\
&\quad + \sum_i q_i \left(\frac{1 - z_i}{2} \right) + C_0.
\end{aligned} \tag{52}$$

Expanding the quadratic terms gives

$$\begin{aligned}
F_Q(\mathbf{z}) &= \frac{1}{4} \sum_{i < j} q_{ij} (1 - z_i - z_j + z_i z_j) \\
&\quad + \frac{1}{2} \sum_i q_i (1 - z_i) + C_0.
\end{aligned} \tag{53}$$

Grouping constant, linear and quadratic spin contributions produces the standard Ising energy function

$$F_Q(\mathbf{z}) = C_{\text{Ising}} + \sum_i h_i z_i + \sum_{i < j} J_{ij} z_i z_j, \tag{54}$$

where

$$J_{ij} = \frac{q_{ij}}{4}, \quad i < j, \tag{55}$$

are the pairwise coupling coefficients and

$$h_i = -\frac{q_i}{2} - \frac{1}{4} \sum_{j \neq i} q_{ij}, \tag{56}$$

are the local magnetic field coefficients associated with each spin.

The constant energy shift is given by

$$C_{\text{Ising}} = C_0 + \frac{1}{4} \sum_{i < j} q_{ij} + \frac{1}{2} \sum_i q_i. \tag{57}$$

Since C_{Ising} is independent of the optimization variables, it merely shifts every energy level by the same amount and therefore has no influence on the optimal solution. Consequently, this constant term is omitted when constructing the quantum Hamiltonian.

4.3 Quantum Cost Hamiltonian

The spin variables obtained in the previous subsection are promoted to quantum operators by replacing each spin variable with the corresponding Pauli-Z operator,

$$z_i \longrightarrow Z_i. \tag{58}$$

The resulting quantum cost Hamiltonian is therefore

$$H_C = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j. \tag{59}$$

Equation (59) constitutes the optimization Hamiltonian used throughout the quantum stage of the proposed framework. The Hamiltonian contains only single-qubit Pauli-Z terms and two-qubit interaction terms, making it directly compatible with gate-based QAOA implementations.

It is important to emphasize that H_C represents only the unconstrained portfolio objective defined in Equation (62). The inequality penalty function,

$$P_{\text{ineq}}(\mathbf{x}),$$

is intentionally excluded from the Hamiltonian construction. As a result, the quantum optimization problem remains significantly smaller than a conventional constrained QUBO formulation that incorporates inequality constraints through slack variables or additional quadratic penalty terms.

The omission of the inequality penalties from the Hamiltonian does not alter the original optimization problem. Instead, these penalties are evaluated during the classical post-processing stage after quantum sampling. Consequently, the role of the quantum computer is to identify promising binary asset-selection configurations, while the enforcement of inequality constraints and the computation of the complete portfolio objective are deferred to the classical optimization stage.

4.4 Quantum Search using CVaR-QAOA

The Ising Hamiltonian in Equation (59) is optimized using the Quantum Approximate Optimization Algorithm (QAOA). Rather than directly solving the complete constrained portfolio optimization problem, the objective of the quantum stage is to efficiently explore the combinatorial search space of binary asset-selection vectors and identify promising candidate portfolios for subsequent classical optimization.

QAOA is a variational hybrid quantum-classical algorithm that alternates between the problem-dependent cost Hamiltonian and a mixing Hamiltonian. Starting from an equal superposition of all computational basis states, the quantum circuit gradually concentrates probability amplitude on low-energy configurations of the cost Hamiltonian through the optimization of a set of variational parameters.

4.4.1 Parameterized Quantum Circuit

For a circuit depth p , the QAOA state is defined as

$$|\psi(\gamma, \beta)\rangle = \prod_{\ell=1}^p e^{-i\beta_{\ell} H_M} e^{-i\gamma_{\ell} H_C} |+\rangle^{\otimes N}, \quad (60)$$

where

- H_C denotes the portfolio cost Hamiltonian defined in Equation (59),
- H_M denotes the mixing Hamiltonian,
- $\gamma = (\gamma_1, \dots, \gamma_p)$ are the cost-Hamiltonian parameters,
- $\beta = (\beta_1, \dots, \beta_p)$ are the mixer parameters.

The mixing Hamiltonian is chosen as

$$H_M = \sum_{i=1}^N X_i, \quad (61)$$

where X_i is the Pauli- X operator acting on qubit i . This Hamiltonian enables transitions between computational basis states, allowing the quantum circuit to explore different asset-selection combinations throughout the optimization process.

4.4.2 CVaR Objective Function

In the proposed hybrid framework, the quantum cost Hamiltonian is constructed only from the unconstrained portfolio objective

$$F_Q(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0, \quad (62)$$

thereby avoiding the introduction of slack variables or additional penalty qubits. The resulting Ising Hamiltonian governs the quantum evolution during the QAOA optimization.

In standard QAOA, the variational parameters are obtained by minimizing the expectation value of the cost Hamiltonian,

$$\langle \psi(\gamma, \beta) | H_C | \psi(\gamma, \beta) \rangle. \quad (63)$$

Instead of optimizing the expectation value, this work employs the Conditional Value-at-Risk (CVaR) objective, which focuses only on the lowest-cost measurement outcomes.

After executing the parameterized quantum circuit, repeated measurements produce a collection of computational basis states,

$$\mathcal{S} = \left\{ \mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(M)} \right\}, \quad (64)$$

where M denotes the total number of measurement shots.

For every sampled bit string $\mathbf{x}^{(s)}$, the corresponding Hamiltonian energy is computed as

$$E_H^{(s)} = \left\langle \mathbf{x}^{(s)} \left| H_C \right| \mathbf{x}^{(s)} \right\rangle. \quad (65)$$

The original inequality constraints are then incorporated through a classical evaluation of the penalty function,

$$P^{(s)} = P_{\text{ineq}} \left(\mathbf{x}^{(s)} \right). \quad (66)$$

The complete objective associated with each sampled portfolio is therefore

$$F^{(s)} = E_H^{(s)} + P^{(s)}. \quad (67)$$

The sampled objective values are sorted in ascending order,

$$F_{(1)} \leq F_{(2)} \leq \dots \leq F_{(M)}. \quad (68)$$

For a confidence parameter $\tau \in (0, 1]$, the CVaR objective is defined as

$$\text{CVaR}_\tau = \frac{1}{M_\tau} \sum_{k=1}^{M_\tau} F_{(k)}, \quad (69)$$

where

$$M_\tau = \lceil \tau M \rceil. \quad (70)$$

The QAOA parameters are obtained by solving

$$(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*) = \arg \min_{\boldsymbol{\gamma}, \boldsymbol{\beta}} \text{CVaR}_\tau. \quad (71)$$

Consequently, the quantum evolution is governed only by the unconstrained Hamiltonian, whereas the classical optimizer is guided by the complete portfolio objective including the inequality penalties.

4.4.3 Generation of Candidate Portfolios

After convergence of the CVaR optimization, the optimal variational parameters $(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)$ are used to execute the optimized QAOA circuit.

Repeated measurements of the optimized circuit generate a collection of candidate binary portfolio selections,

$$\mathcal{S} = \left\{ \mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(M)} \right\}, \quad (72)$$

which represent promising asset combinations identified by the quantum search. These binary portfolios are subsequently forwarded to the classical optimization stage for continuous portfolio weight allocation.

4.5 Continuous Portfolio Weight Optimization

The binary decision vectors generated by the optimized QAOA circuit specify only the subset of assets selected for investment. They do not determine the continuous proportion of capital allocated to each selected asset.

Therefore, for every candidate portfolio obtained from the quantum stage, a classical optimization problem is solved to determine the optimal portfolio weights.

The optimization problem is formulated as

$$\begin{aligned} \min_{\mathbf{w}} \quad & -\alpha \boldsymbol{\mu}^T \mathbf{w} - \beta \mathbf{y}^T \mathbf{w} + \lambda \mathbf{w}^T \Sigma \mathbf{w} \\ & + \gamma \mathbf{d}^T \mathbf{w} + \delta \mathbf{c}^T \mathbf{w} \end{aligned} \quad (73)$$

subject to

$$\sum_{i=1}^N w_i = 1, \quad (74)$$

$$w_i \geq 0, \quad (75)$$

$$w_i = 0, \quad \text{if } x_i = 0, \quad (76)$$

together with any additional portfolio constraints.

The quantum stage therefore performs the combinatorial asset-selection task, whereas the classical optimizer computes the optimal continuous capital allocation among the selected assets.

4.6 Overall Hybrid Optimization Algorithm

Hybrid CVaR-QAOA Portfolio Optimization:

1. Collect historical market data.
2. Compute expected returns, covariance matrix, growth income, dividend yield and transaction costs.
3. Construct the portfolio objective function.
4. Separate the objective into

$$F_Q(\mathbf{x}) = Q(\mathbf{x}) + L(\mathbf{x}) + C_0$$

and

$$P_{\text{ineq}}(\mathbf{x}).$$

5. Construct the Ising Hamiltonian H_C from $F_Q(\mathbf{x})$.
6. Initialize the QAOA parameters $(\boldsymbol{\gamma}, \boldsymbol{\beta})$.
7. Execute the QAOA circuit and obtain M measurement samples.
8. Compute the Hamiltonian energy $E_H^{(s)}$ for every sampled bit string.
9. Evaluate the inequality penalty $P_{\text{ineq}}(\mathbf{x}^{(s)})$ for every sampled portfolio.
10. Compute

$$F^{(s)} = E_H^{(s)} + P_{\text{ineq}}(\mathbf{x}^{(s)}).$$

11. Compute the CVaR $_{\tau}$ objective using the lowest-cost samples.
12. Update $(\boldsymbol{\gamma}, \boldsymbol{\beta})$ using a classical optimizer.

13. Repeat Steps 7–12 until convergence.
14. Execute the optimized QAOA circuit to generate candidate portfolios.
15. Compute the continuous portfolio weights using a classical optimizer.
16. Return the final optimized portfolio.

5 Experimental Results

The proposed portfolio optimization framework was evaluated using a universe of 20 assets, from which 12 assets were selected for the experimental analysis. The cardinality parameter was set to $K = 10$, meaning that the final portfolio was required to select exactly 10 assets. Both the classical MIQP formulation and the hybrid CVaR-QAOA formulation were evaluated using the same financial inputs, portfolio constraints, and investment preferences.

The classical optimization problem was solved using the **Gurobi** optimizer. For the hybrid quantum-classical approach, the binary asset-selection problem was solved using **hybrid CVaR-QAOA**, while the QAOA variational parameters (γ, β) were optimized using the **Powell** classical optimization algorithm. The resulting quantum candidate portfolio was subsequently evaluated and assigned continuous portfolio weights using the classical optimization stage.

5.1 Classical and Hybrid Optimization Results

Both approaches produced feasible portfolios with zero hard-constraint violations. The classical Gurobi solution achieved an objective value of 82.361701, while the hybrid CVaR-QAOA solution achieved a slightly lower objective value of 82.275073.

Table 1: Comparison of Classical Gurobi and Hybrid CVaR-QAOA portfolios.

Metric	Classical	Hybrid CVaR-QAOA
Expected Return	15.13%	15.18%
Risk / Volatility	7.99%	7.97%
Drawdown	9.81%	9.84%
Transaction Cost	0.10%	0.10%
Objective Value	82.361701	82.275073
Constraint Breaches	0	0
Selected Assets	10	10

The selected assets and their portfolio weights are presented in Table 2. Both methods selected the same dominant positions in TLT, VNQ, and XLE. The only difference in asset selection was between SPY and QQQ.

Table 2: Asset selection and portfolio weights.

Asset	Classical	CVaR-QAOA	Selection
TLT	40%	40%	Both
VNQ	40%	40%	Both
XLE	13%	13%	Both
SLV	1%	1%	Both
USO	1%	1%	Both
XLRE	1%	1%	Both
DBC	1%	1%	Both
REET	1%	1%	Both
XLK	1%	1%	Both
SPY	1%	0%	Classical only
QQQ	0%	1%	Quantum only

5.2 CVaR-QAOA Configuration

The CVaR-QAOA experiment was performed with a circuit depth of $p = 1$ and a CVaR confidence level of $\tau = 0.10$. The QAOA parameter optimization used 256 measurement shots per iteration and a maximum of 30 classical optimization iterations. After convergence, 2048 final measurement shots were used to generate the candidate portfolio solutions. The cardinality penalty coefficient was set to $\eta_2 = 60.0$. The optimized variational parameters were $\gamma = 3.49613347$ and $\beta = 1.94161104$, resulting in a CVaR value of 32.279539 and a Hamiltonian energy of -63.227640 .

The two portfolios have nine assets in common, corresponding to a selection agreement of 81.82%. The classical solution selected SPY, whereas the hybrid CVaR-QAOA solution selected QQQ. Apart from this substitution, the portfolio allocations were identical.

The hybrid approach required approximately 0.31 seconds for the quantum optimization stage. The resulting CVaR value was 32.279539, and the Hamiltonian energy of the selected solution was -63.227640 .

5.3 Discussion

The classical Gurobi and hybrid CVaR-QAOA approaches produced very similar portfolios, with both satisfying all hard constraints. The hybrid approach achieved a slightly higher expected return (15.18% vs. 15.13%) and slightly lower risk (7.97% vs. 7.99%). The objective values were also very close, with a difference of approximately 0.11%. Both approaches selected the same nine assets, differing only between SPY and QQQ, resulting in an 81.82% selection agreement. Therefore, the results indicate that hybrid CVaR-QAOA was able to identify a portfolio competitive with the classical Gurobi solution while satisfying all hard constraints.

Note. The full potential of the hybrid CVaR-QAOA approach has not been explored in this experiment. The current results are based on $p = 1$, a fixed CVaR level of $\tau = 0.10$, and the selected optimization settings. Higher circuit depths, different CVaR levels, more measurement shots, and execution on quantum hardware may produce different results. Therefore, the current results show that the hybrid approach is competitive with the classical solution, but they do not represent the maximum performance that hybrid CVaR-QAOA can achieve.

6 Interactive Portfolio Co-Pilot

A Streamlit-based interactive Portfolio Co-Pilot was developed to provide a user-facing interface to the proposed portfolio optimization framework. The application allows users to adjust investment preferences, including growth, income, drawdown control, cost sensitivity, and risk aversion. It then runs the classical MIQP and hybrid CVaR-QAOA optimization workflows, validates hard-constraint feasibility, compares the resulting portfolios, and provides a recommendation with an explanation of the associated trade-offs.

The current Co-Pilot serves as a research and demonstration interface. Future extensions may incorporate dynamic market data, additional portfolio constraints, scenario and sensitivity analysis, and direct quantum-hardware execution.

Figure 1 shows the initial view of the Portfolio Co-Pilot dashboard, where the user can configure the optimization parameters through an interactive interface. The tunable parameters can be adjusted using the provided sliders according to the user’s requirements. Once the desired parameters are selected, the user can click the **Run Optimization** button to initiate the optimization process. The interactive terminal then executes the optimization loop and displays the progress and intermediate information during execution. Upon completion, the optimized portfolio and associated performance results are presented through the dashboard, as shown in Figure 2. The generated results are also made available for download, allowing the user to further analyze or store the optimization outputs for subsequent use.

9 Project Links

GitHub Repository: [Multi-Asset Portfolio Construction](#)

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